

The National Higher School of Artificial Intelligence
Introduction to AI
QUIZ #2 11/05/2023
Duration: 25 minutes

QUESTIONS:

1. Which of the following are the two key characteristics of Genetic Algorithms?
 - a) **The crossover operation and the fitness function**
 - b) Random mutation and Crossover
 - c) Random mutation and the individuals in the population
 - d) Random mutation and the fitness function
 - e) None of the above
2. For CSPs, which of the following is FALSE?
 - a) **The *Minimum Remaining Values* strategy is better at finding the best value from a domain to start with for a given variable than the *Least Constraining Value*.**
 - b) The *Degree Heuristic* can be used when there is a tie in terms of the selection of a variable using another heuristic that is used for the selection of a variable in the partial assignment being built.
 - c) The *Least Constraining Value heuristic* tries to leave the maximum flexibility for subsequent variable assignments.
 - d) If we try to find all the solutions to a CSP, then the ordering of the variables and the domain values for each of these variables does not matter.
 - e) None of the above is False.
3. In many problems the path to the goal is irrelevant. For this class of problems, which of the following techniques can be used to solve them?
 - a. Informed Search
 - b. Uninformed Search
 - c. **Local Search**
 - d. both a and b
 - e. None of the above
4. Hill climbing is commonly known assearch because it selects a suitable neighbour state without being thoughtful about where to go next beyond this choice.
 - a. Needy local search
 - b. Random-restart local search
 - c. **Greedy local search**
 - d. Optimal local search
 - e. None of the above
5. Which of the following algorithms keeps track of k states instead of just one.
 - a. Hill-Climbing search
 - b. **Local Beam search**
 - c. Stochastic hill-climbing search
 - d. Random restart hill-climbing search
 - e. None of the above

6. Which of the following statements is FALSE about alpha-beta pruning?
- Alpha-Beta pruning returns the same move as *minimax* would.
 - If at any MAX node, the backed-up value is such that α is strictly less than β , Alpha-Beta pruning has a chance to return a better value than *minimax* up to the root of the game tree for the MAX player.**
 - Minimax can be applied to trees of any depth.
 - Alpha-beta search updates the values of β as it goes along and prunes the remaining branches at a node as soon as the value of the current node is known to be worse than the current β value for MIN.
 - The effectiveness of alpha-beta pruning depends on the order in which the states are examined.
7. Which of the following is FALSE about Constraint Satisfaction Problems?
- Alldiff* problems are problems where all variables must have distinct values except the ones defined as exceptions in the CSP formulation.**
 - One can detect inconsistency with respect to the *Atmost* global constraint by checking the sum of the minimum values of the current domains of the variables.
 - Whenever the problem is such that it is not possible to represent the domain of some variables as large sets of integers, the use of upper and lower bounds for the representation of the domains of these variables can be exploited.
 - Sudoku can be considered as an *Alldiff* CSP.
 - Many CSPs cannot be solved by inference alone; and search strategy can then be needed to find a solution.
8. Which of the following are the main disadvantages of a hill-climbing search?
- It stops at a local optimum and does not find the optimum solution**
 - It stops at a global optimum and does not find the optimum solution
 - It does not find the optimum solution and does a lot of searching for a solution
 - It fails to find a solution
 - None of the above
9. Local search algorithms are not systematic. Which of the following, is their most important advantage?
- They require less memory
 - They require more time
 - They search for a solution in a big infinite space
 - They require less memory and search for a solution in a big infinite space**
 - None of the above
10. A zero-sum game is a game.
- Single Player
 - Two-Player
 - Three-Player
 - Multiplayer**
 - None of the above

11. Where do the values in alpha-beta search get updated?

- a. **Along the path of the search**
- b. At the initial state itself
- c. At the terminal state of any path of the search
- d. None of the above

12. What is called a transposition table?

- a. The set of all situations where the MAX and MIN players “ate” pieces of each other as an attack and an immediate counter-attack.
- b. A hash table of the next positions to be seen and their associated values
- c. **A hash table of the of previously seen positions and their associated values**
- d. The previous and the next values of the state in the search
- e. None of the above

13. Which of the following statements is FALSE?

- a) An inference algorithm i is sound if whenever $KB \vdash_i \alpha$ it is also true that $KB \models \alpha$
- b) An inference algorithm is complete if whenever $KB \models \alpha$ it is also true that $KB \vdash_i \alpha$
- c) Proving α from KB by checking the unsatisfiability of $(KB \wedge \neg \alpha)$ corresponds exactly to a proof by contradiction.
- d) **Although Resolution taken on its own as an inference rule is less effective than the union of all the other inference rules (Modus Ponens, And-elimination, logical equivalences, etc.), the complexity of resolution makes it more interesting to use in all theorem proving with propositional logic.**
- e) There is a single statement which is false among the suggested answers to this question.

14. Which function is used to calculate the terminal nodes in a game tree?

- a. Transposition function
- b. **Evaluation function**
- c. Alpha-beta pruning
- d. Only the answers a and c above
- e. None of a, b, c.

15. Which search is similar to minimax search?

- a. Hill-climbing search
- b. **Depth-first search**
- c. Breadth-first search
- d. All of the above
- e. None of a, b, c.

16. The initial state and the legal moves for each side define the _____ for the game.

- a. Search tree
- b. **Game tree**
- c. State Space Search
- d. A Random forest
- e. None of the above

17. Which search is equal to minimax search but eliminates the branches that can't influence the final decision?
- a. Depth-first search
 - b. Breadth-first search
 - c. **Alpha-beta pruning**
 - d. None of the above
 - e. All of a, b, and c.
18. In game playing, which is a possible approach to forward pruning?
- a) Hill Climbing
 - b) Bidirectional search
 - c) Stochastic Hill Climbing.
 - d) Constraint Satisfaction Problem
 - e) **Beam search**
19. To which depth can alpha-beta pruning be applied?
- a. maximum 10 states
 - b. maximum 8 States
 - c. maximum 6 States
 - d. **Any depth**
 - e. Depending on the problem
20. An inference algorithm is complete only if
- a. It can derive any sentence
 - b. **It can derive any sentence that is entailed**
 - c. It is truth-preserving
 - d. It can derive some sentence that is entailed
 - e. None of the above
21. The term _____ is used for a depth-first search that chooses values for one variable at a time and returns when a variable has no legal values left to assign.
- a) Forward search
 - b) **Backtracking search**
 - c) Down-Hill algorithm
 - d) Reverse-Down-Hill search
 - e) None of the above
22. Constraint satisfaction problems on finite domains are typically solved using a form of _____
- a. Search Algorithms
 - b. Heuristic Search Algorithms
 - c. Greedy Search Algorithms
 - d. **All of the above**
 - e. None of the above
23. Which of the following algorithms does not maintain a search tree?

- a. Depth-limited search
- b. A*-search
- c. Genetic Algorithm**
- d. All of the above
- e. None of the above

24. An inference algorithm is sound only if

- a. It can derive any sentence
- b. It can derive any sentence that is entailed
- c. It is truth-preserving**
- d. It can derive some sentence that is entailed
- e. None of the above

25. Which of the following statements is TRUE when using a simulated annealing (SA) algorithm?

- a) SA is like Hill Climbing but instead of picking a random move, it selects the best one.
- b) If a move that is being considered improves the current state, it is always accepted with a probability p and not selected with a probability $1-p$.
- c) The probability of selecting a move increases exponentially with the “badness” of the move—the amount ΔE by which the evaluation is worsened.
- d) The probability of selecting a move decreases as the “temperature” T goes down.**
- e) If the schedule decreases T slowly enough, the algorithm will find a global optimum with probability equal to 1.

26. In propositional logic, suppose you have two propositions Stench23 and Stench44 meaning respectively that there is Stench in square [2,3] and square [4,4]. Which of the following is True.

- a) The two propositions are completely unrelated.**
- b) The sentence $\text{Stench23} \wedge \text{Stench44}$ is unsatisfiable.
- c) Either Stench23 is True and Stench44 is False, or vice-versa.
- d) The sentence $\text{Stench23} \wedge \text{Stench44}$ is valid.
- e) None of the above.